

# Ashfaque Thonikkadavan

Forecasting & optimization consultant. I build models that turn uncertain demand into concrete decisions — inventory levels, replenishment, staffing and resource allocation.

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## 01 PROFILE

A **PhD-trained researcher** working in applied forecasting and operations research. My background is computational physics — large numerical systems, simulation, statistical analysis — and I bring that rigour to business problems. Most forecasting work stops at accuracy metrics; I care about the last mile, whether a forecast actually reduces stockouts, holding cost or idle capacity downstream.

## 02 WHAT I DO

|                                                                                                            |                                                                                                                               |                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 01                                                                                                         | 02                                                                                                                            | 03                                                                                                                               |
| <b>Demand forecasting</b>                                                                                  | <b>Inventory &amp; planning</b>                                                                                               | <b>Constrained optimisation</b>                                                                                                  |
| Demand forecasts with honest error estimates, built on walk-forward backtesting rather than random splits. | Turning forecast error into safety stock, order-up-to levels and replenishment policy, scored on realised cost and fill rate. | Mixed-integer models for allocating scarce resources under budget, capacity and policy constraints, solved to proven optimality. |

## 03 HOW I WORK

|                                                                               |                                                                                    |                                                                                       |                                                                              |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 01                                                                            | 02                                                                                 | 03                                                                                    | 04                                                                           |
| <b>Frame the decision</b>                                                     | <b>Baseline first</b>                                                              | <b>Validate honestly</b>                                                              | <b>Report in business units</b>                                              |
| What decision are we actually trying to improve, and what does it cost today? | Establish the cheapest model that works before adding complexity or buying a tool. | Walk-forward backtests, blind holdouts, and explicit failure modes. No random splits. | Cost, service level, allocation, risk — plus what the model still cannot do. |

## 04 SELECTED WORK

Three representative systems

### Retail demand forecasting → inventory decisions

CASE STUDY / FORECASTING + INVENTORY

Walmart M5 dataset · store CA\_1 · 300 SKUs · 6 models · 28-day holdout · 2026

|          |                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROBLEM  | A forecast is only worth having if it changes a decision downstream. Which demand model is accurate — and which one actually lowers inventory cost?                                                                                                                                                                                                                                                            |
| APPROACH | Causally-safe feature engineering (lags, rolling statistics, calendar, events, price) feeding recursive and direct LightGBM models, benchmarked against five operational baselines including seasonal naive, moving averages and Croston. The selected forecast drives a periodic-review replenishment simulation: risk-period demand, error-based safety stock, order-up-to level, holding and stockout cost. |

VALIDATION

Six walk-forward backtest windows for selection, then one blind 28-day holdout on unseen data. Every candidate is replayed through an identical replenishment policy, so any cost difference belongs to the forecast alone.

OUTCOME

The selected forecast was the cheapest of all six under that fixed policy — \$16,479 against \$18,081 for the seasonal moving average — and is served through PostgreSQL and a FastAPI endpoint rather than a notebook.

|                            |                             |                           |                       |
|----------------------------|-----------------------------|---------------------------|-----------------------|
| 8.9%                       | -\$1,601                    | 90.4%                     | 1.09                  |
| LOWER TOTAL INVENTORY COST | VS. SEASONAL MOVING AVERAGE | FILL RATE, 28-DAY HOLDOUT | MAE UNITS / SKU / DAY |

### Optimising resource allocation under constraints

CASE STUDY / OPTIMISATION

Mixed-integer linear programming · SoFIFA 25 dataset · live Streamlit app · 2025

|            |                                                                                                                                                                                                                                                                                                                                                                  |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROBLEM    | Choosing a team under a wage budget, a fixed formation, positional hierarchies and locked-in selections is a constrained selection problem. Heuristic scoring gives a plausible answer, never a provably best one.                                                                                                                                               |
| APPROACH   | Formulated as a MILP with role-aware binary variables $x[\text{player}, \text{role}]$ , solved to optimality with PuLP. Sixteen roles are modelled, a player fills at most one, and budget, squad size, formation balance, age window and locked players all enter as hard constraints. A natural-language layer turns plain-text requirements into constraints. |
| VALIDATION | Every solution is checked for constraint satisfaction and re-solved across scenario variations, so infeasible selections surface to the user rather than returning silently.                                                                                                                                                                                     |
| OUTCOME    | A working, publicly reachable application rather than a research script. The same pattern transfers directly to staffing, scheduling and capital allocation.                                                                                                                                                                                                     |

|                                |                           |                              |                                |
|--------------------------------|---------------------------|------------------------------|--------------------------------|
| Optimal                        | 16                        | 6                            | Live                           |
| SOLUTION STATUS, NOT HEURISTIC | ROLES MODELLED PER PLAYER | CONSTRAINT FAMILIES ENFORCED | DEPLOYED STREAMLIT APPLICATION |

### Production ML under uncertainty

CASE STUDY / PRODUCTION ML

800k records · severe class imbalance · FastAPI + Docker · 2025

|            |                                                                                                                                                                                                                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROBLEM    | A content-classification service facing a <10% minority class and noisy, out-of-distribution traffic. A single hard threshold either floods the review queue or silently drops real violations.                                                                                                                              |
| APPROACH   | TF-IDF + linear SVM with Platt scaling for probability calibration, wrapped in a three-tier confidence router: high-confidence predictions are auto-handled, low-confidence ones escalate to review, the ambiguous middle band is logged for retraining. Containerised as a FastAPI service with structured request logging. |
| VALIDATION | Recall deliberately optimised over precision, because the cost of a missed violation outweighs a false positive. Load behaviour stress-tested separately from model accuracy.                                                                                                                                                |
| OUTCOME    | A service that degrades predictably under distribution shift instead of failing silently — and generates its own training signal from the escalation queue.                                                                                                                                                                  |

|                   |                          |           |                          |
|-------------------|--------------------------|-----------|--------------------------|
| 800k              | 85%                      | 61%       | 3                        |
| RECORDS PROCESSED | RECALL ON MINORITY CLASS | PRECISION | CONFIDENCE ROUTING TIERS |

## Scientific Researcher

Oct 2022 – Oct 2025

IPCMS, University of Strasbourg · CNRS

- Ran large-scale finite-element micromagnetic simulations of topologically complex structures, analysing **10M+ data points** across high-dimensional 3D meshes.
- Built automated Python pipelines for extraction, transformation and analysis of simulation output, **cutting analysis time by roughly 60%** across repeated runs.

### 06 EDUCATION & RESEARCH

## PhD, Condensed Matter Physics

2022 – 2025

IPCMS & University of Strasbourg, France

- Dissertation on spin-wave dynamics in topologically complex magnetic structures. Trained in large-scale computation, algorithm optimisation and research-grade data pipelines.

## MSc, Physics

2018 – 2020

Cochin University of Science & Technology, India · statistical mechanics, computational physics, numerical methods

## BSc, Physics

2015 – 2018

University of Calicut, India

### 07 PUBLICATION & WRITING

- **Rotating spin wave modes in nanoscale Möbius strips.** *NPJ Spintronics*, 4(4), 2026. [doi.org/10.1038/s44306-025-00118-8](https://doi.org/10.1038/s44306-025-00118-8)
- Technical writing on forecasting, optimisation and the research-to-industry transition — [medium.com/@cmtwskb](https://medium.com/@cmtwskb)

### 08 TOOLKIT

|                        |                                                                                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| LANGUAGES              | Python, SQL, LaTeX · English (fluent), French (learning)                                                                         |
| FORECASTING            | Walk-forward backtesting, ARIMA, SARIMAX, exponential smoothing, Prophet, LightGBM, WRMSSE                                       |
| OPTIMISATION           | MILP (PuLP), constraint programming, OR-Tools, operations research                                                               |
| ML, STATS & SCIENTIFIC | scikit-learn, XGBoost, probability calibration, imbalanced classification, PCA, NumPy, SciPy, pandas, FFT, spatial interpolation |
| DELIVERY               | FastAPI, Docker, PostgreSQL, Streamlit, Git, Linux, Power BI                                                                     |

### WORKING TOGETHER

I take on a few engagements at a time — usually a forecasting or optimisation problem where the data already exists and the impact is measurable. A first conversation is usually enough to tell whether I can help.

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